CUSTOMER CHURN INTERVENTION PLAYBOOK

Telco Customer Churn Analysis Project

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Tools Used: Python | MySQL | Power BI

1. Project Overview

This playbook is the output of an end-to-end customer churn analysis performed on a Telco dataset of 7,043 customers. The goal is to identify why customers are leaving and define specific data-driven actions to reduce churn.

Metric	Value
Total Customers Analyzed	7,043
Total Churned Customers	1,869
Overall Churn Rate	26.5%
Active Customers	5,174
Avg Monthly Charges (Churned)	~$74/month
Avg Monthly Charges (Active)	~$61/month
Analysis Tools	Python, MySQL, Power BI

2. Key Findings — Root Causes of Churn

Finding 1 — Contract Type is the #1 Churn Driver

Evidence:

- Month-to-month customers churn at 42%
- One-year contract customers churn at 11%

- Two-year contract customers churn at only 3%

Root Cause:

No long-term commitment means zero switching cost. Customers can leave at any time with no financial penalty, making them highly vulnerable to competitor offers.

Finding 2 — New Customers Churn the Most (Broken Onboarding)

Evidence:

- Customers in the 0-12 month tenure group churn at ~47%
- Churn drops significantly after 24 months of tenure

Root Cause:

Customers do not see enough value in the early stages. Expectations set during the sales process are not being met by the actual service experience.

Finding 3 — Fiber Optic Pricing vs Quality Gap

Evidence:

- Fiber optic customers churn at 42% — highest of all service types
- These customers pay the highest monthly charges

Root Cause:

Premium-priced customers expect premium-quality service. When the experience does not justify the price, they switch to competitors offering better value.

Finding 4 — Low Feature Engagement = High Churn Risk

Evidence:

- Customers with no TechSupport churn at 42% vs 15% for users with support
- Customers with no OnlineSecurity churn at 42% vs 15% for users with security

Root Cause:

Low product engagement means low switching cost. Customers who do not use features do not feel dependent on the product, making it easy to cancel.

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3. Churn Intervention Rules

Rule 1 — High Risk Customers

Field	Details
Trigger Condition	tenure < 12 months AND Contract = Month-to-month

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Who They Are	New customers with no long-term commitment
Churn Rate	~47% — highest risk group
Priority	URGENT — Action within 24 hours
Customer Count	~1,000+ customers flagged

Recommended Actions:

- Send personalised retention email within 24 hours of flagging
- Offer 20% discount on upgrade to annual contract
- Assign dedicated onboarding specialist for first 30 days
- Schedule a check-in call at Day 7, Day 30, and Day 90

Expected Impact:

Reduce new customer churn rate from 47% to below 30% within 6 months.

Rule 2 — High Value Customers Lost

Field	Details
Trigger Condition	MonthlyCharges > $89.85 AND Churn = Yes
Who They Are	Premium customers who have already churned
Churn Rate	High revenue impact segment
Priority	HIGH — Action within 48 hours
Customer Count	~400+ customers flagged

Recommended Actions:

- Assign dedicated account manager immediately
- Send personalised win-back offer within 48 hours
- Offer 3 months free premium service as re-engagement incentive
- Conduct exit interview to understand specific reason for leaving
- Flag account for quarterly review to prevent future churn

Expected Impact:

Recover 10-15% of lost high value customers through win-back campaigns.

Rule 3 — Feature Gap Customers

Field	Details
Trigger Condition	TechSupport = No AND OnlineSecurity = No

Who They Are	Customers not using key product features
Churn Rate	~42% vs 15% for feature users
Priority	MEDIUM — Action within 1 week
Customer Count	~2,000+ customers flagged

Recommended Actions:

- Send feature education email series over 4 weeks
- Offer free 30-day trial of TechSupport + OnlineSecurity bundle
- Add in-app notification highlighting security and support benefits
- Train customer service team to mention features during every call

Expected Impact:

Increase feature adoption by 25% and reduce churn in this segment by 20%.

Rule 4 — Fiber Optic Dissatisfied Customers

Field	Details
Trigger Condition	InternetService = Fiber optic AND tenure < 24 months
Who They Are	Premium internet customers churning early
Churn Rate	~42% — highest of all service types
Priority	MEDIUM — Action within 1 week
Customer Count	Fiber optic segment customers

Recommended Actions:

- Proactive service quality check call within first week of flagging
- Offer complimentary speed upgrade at no extra cost
- Send customer satisfaction survey after every service interaction
- Review fiber optic pricing vs competitor benchmarks quarterly

Expected Impact:

Reduce fiber optic churn rate from 42% to below 28% within 6 months.

4. Intervention Priority Matrix

Customer	Churn Rate	Priority	Action	Owner

Segment			Timeline	
High Risk (New + Month-to-month)	~47%	URGENT	Within 24 hours	Retention Team
High Value Lost (Churned)	Revenue Impact	HIGH	Within 48 hours	Account Manager
Feature Gap Customers	~42%	MEDIUM	Within 1 week	Customer Success
Fiber Optic Dissatisfied	~42%	MEDIUM	Within 1 week	Technical Team
Stable Customers	<10%	LOW	Monthly check-in	Automated

5. Expected Business Impact

Metric	Current	Target (6 months)
Overall Churn Rate	26.5%	Below 15%
New Customer Churn (0-12m)	~47%	Below 30%
Fiber Optic Churn	~42%	Below 28%
Feature Gap Churn	~42%	Below 25%
Customers at Risk	1,869 churned	Recover ~560 (30%)

6. Tools & Methodology

Tool	Purpose
Python (Pandas, Seaborn, Matplotlib)	Data cleaning, EDA, visualizations
MySQL Workbench	SQL queries, churn segmentation, playbook triggers
Power BI Desktop	3-page interactive dashboard with KPIs and slicers

Kaggle Dataset	Telco Customer Churn — 7,043 customers, 21 columns